R Programming Lab 05

Social Network Analysis Using R

Name	Tanisha Gupta
Roll Number	23102A0002

1. Aim

To create, visualize, and analyse networks using R, calculate network measures, identify important nodes in a directed network, and detect communities in a larger network dataset.

2. Tools and Dataset

The experiment uses the **igraph** package for graph creation, network visualization, centrality measures, hub and authority scores, and community detection. It first demonstrates concepts with small example networks, then imports a network dataset from a CSV file containing pairs of connected nodes.

3. Small Network Analysis

An undirected graph and a directed graph are created to demonstrate different connection types. For the directed example containing Amy, Ram, Li, and Kate, the calculated degree values are shown below.

Node	In-degree	Out-degree	Total degree
Amy	1	2	3
Ram	1	1	2
Li	3	1	4
Kate	0	1	1

The example network has a diameter of 2, edge density of approximately 0.4167, and reciprocity of 0.4. Closeness and betweenness centrality are also calculated to compare the structural importance of nodes.

4. Analysis of the Larger Network

The CSV data is converted into a directed graph. The resulting network contains 52 nodes and 290 edges. A degree histogram and network plots are generated to explore the distribution of connections and the overall structure. Node sizes are adjusted according to degree in one visualization.

5. Hub and Authority Scores

Hub scores identify nodes that point to important authorities, while authority scores identify nodes receiving links from important hubs. The top scores from the executed analysis are summarized below.

Rank	Top hub	Hub score	Top authority	Authority score
1	CC	1.0000	CA	1.0000

Rank	Top hub	Hub score	Top authority	Authority score
2	CB	0.7146	DF	0.1926
3	CD	0.2350	DD	0.1410
4	BC	0.2322	CC	0.1122
5	BE	0.2195	BE	0.1095

Node CA has the highest overall degree and the highest authority score, while node CC has the highest hub score.

6. Community Detection

The directed network is converted to an undirected graph for community detection using the edge-betweenness method. The executed analysis identifies 16 communities and a modularity score of 0.2571. This indicates some community structure, although the score alone does not establish the practical meaning of each group.

7. Summary of Results

Measure	Result
Nodes in imported network	52
Edges in imported network	290
Highest-degree node	CA
Highest hub-score node	CC
Highest authority-score node	CA
Communities detected	16
Modularity	0.2571

8. Conclusion

This experiment demonstrates how R can be used to build and visualize directed and undirected networks, calculate measures of connectivity and centrality, rank hubs and authorities, and identify communities. The larger network analysis highlights node CA as the most connected and highest-authority node, node CC as the strongest hub, and 16 detected communities. These measures provide a structured way to explore network organization and node importance.

Note: The community analysis uses an undirected version of the imported graph, so direction is not considered during community detection.